

ARIN PAUL

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EDUCATION

University of Southern California

Master of Applied Data Science (MADS) | CGPA – 4.00/4.00

Coursework – Foundations of Data Management, Machine Learning, Information Retrieval & Search Engines

Los Angeles, CA

May 2027

Vellore Institute of Technology

B. Tech, Electronics and Communication Engineering | CGPA – 9.25/10

Vellore, India

May 2022

SKILLS

Languages & Tools: Python, Java, SQL, REST, MongoDB, PySpark, Docker, Kubernetes, Helm, Git, AWS, FAISS

Libraries and Frameworks: PyTorch, NLTK, Hugging Face, NumPy, Pandas, TensorFlow, Scikit-Learn, OpenCV, FastAPI

Technologies: Machine Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Deep Learning Systems, Computer Vision, RAG, Cloud Computing, Optimization, CI/CD, A/B Testing, Multimodal Models, Prompt Engineering

EXPERIENCE

Moneyview Limited

Software Engineer – 2

Software Engineer – 1

Bengaluru, India

Oct 2024 – Jul 2025

Jul 2022 – Sep 2024

- Redesigned microservices release process with Jenkins, **blue-green** deployment strategy and multi-stage Dockerfile, **cutting deployment turnaround by 15%**, increasing deployment frequency to 3 times per week, and **reducing change failure rate to 5%** over 6 months
- Designed and implemented Infrastructure as Code (IaC) with **Terraform** for the entire organization, improving consistency, scalability, and management of cloud infrastructure by automating deployments and enabling version control
- Migrated pre-production environments from the production **AWS** account for the entire development team as part of the **prod-isolation objective**, fortifying security by minimizing cross-environment contamination risks and **lowering mean time to recovery for test failures by 40%** via safer rollbacks
- Spearheaded **DevSecOps** practices by integrating vulnerability scanning, secrets management, and compliance checks into **CI/CD** pipelines, leveraging pre-commit hooks and **AWS Secrets Manager**
- Partnered with development and QA teams to streamline environment configurations and overhaul quality assurance processes, **trimming JIRA ticket volume by 25%**

Moneyview Limited

Software Engineer Intern

Bengaluru, India

Mar 2022 – Jun 2022

- Migrated **12** applications from **EC2 static servers to an auto-scaling group**, facilitating dynamic scaling based on load and decreasing deployment time by **17%**
- Enhanced security of cloud infrastructure by transitioning over **200 users** from Identity and Access Management (IAM) to Single Sign-On (SSO) and implementing Just-in-Time (JIT) access for database resources

PROJECTS

University of Southern California

AMIE - Automated Meeting Insight Extraction using Transformer-Based NLP

Los Angeles, CA

Apr 2026

- Developed an end-to-end transformer-based NLP pipeline (BART, RoBERTa, BERTopic) processing meeting audio via OpenAI Whisper and pyannote diarization, achieving **ROUGE-2 score of 0.42** on summarization and **91% accuracy** on action item extraction across multi-speaker transcripts.
- Fine-tuned RoBERTa for per-speaker sentiment classification and trained a custom spaCy NER model to extract people, organizations, and deadlines, achieving an **F1-score of 0.88** on entity extraction and **89% sentiment accuracy** with neuralcoref-based coreference resolution to attribute action items to the correct speaker.
- Benchmarked and optimized AMIE's summarization models by comparing BART, T5, and TextRank across ROUGE-1, ROUGE-2, and BERTScore, selecting BART as the best-performing model with a 17% improvement in ROUGE-2 over the TextRank baseline, and implemented hierarchical summarization for transcripts beyond the 1,024-token model context limit.

LogMind - Log Intelligence & Autonomous Incident Response System

Feb 2026

- Architected a log intelligence platform, integrating **ELK Stack**, **RAG pipeline (LangChain + FAISS)**, and a **fine-tuned Llama-3.1-8B**, enabling automated root cause analysis by mapping production errors to the exact function in the codebase.
- Fine-tuned **Llama-3.1-8B via QLoRA** on domain-specific log-error-fix pairs, achieving **increased 18% root cause accuracy** and **+0.11 ROUGE-L** over the base model baseline, indexing **10M+ log entries** and **500K+ lines of source code** into a **FAISS index** using **sentence transformers (all-MiniLM-L6-v2)**
- Engineered a **5-tool MCP action layer** enabling a fully autonomous incident loop: anomaly detection → root cause reasoning → automated Jira ticket generation → real-time Slack alerting → escalation handling. Eliminating manual handoffs across monitoring, debugging, and escalation with **zero human intervention**

Real-Time Emergency Vehicle Priority System using Multimodal Deep Learning

- Built a real-time ambulance detection system using **YOLOv8**, paired with a **ResNet-based** audio classifier on Mel spectrograms, achieving 92.37% training and 88.72% test accuracy with an **F1 score of 0.89**
- Engineered a dual-modal deep learning pipeline combining YOLOv8 with attention-based feature extraction for visual detection and a transformer-based audio classifier leveraging **Mel spectrogram** embeddings using Focal Loss, Layer Normalization, and GELU activations to confirm emergency vehicle presence with **88.72% accuracy**